Phase pair and ellipse predictors for lossless compression

A retrospective evidence study of complete archive size, exact reconstruction, and transfer

Abstract

Structured prediction can reduce a lossless archive only when the saved residual bytes exceed the cost of the structure used to predict them. This study examines three related but distinct implementations: a phase-pair CSV mode, an operator-informed ellipse predictor for simulation response tables, and a causal oscillator predictor for audio. Historical complete-archive measurements show a 7.6861% reduction on development CSV data, no change across 60 separate CSV controls, and a 16.0736% reduction on observation CSVs exported from one independent simulated pendulum dataset. On four Remembrance response tables, the ellipse archive totals 537,195 bytes against 560,859 for its standard baseline. A separate audio adaptation improves on fixed second differences but loses to unrestricted two-lag prediction, FLAC, and WavPack.

The evidence supports narrow, data-dependent compression gains, not a universal advantage for geometric predictors. Exact signs, numerical residuals, token spelling, model parameters, and framing remain part of the stored information. Archive sizes are recovered experimental measurements, not newly encoded benchmarks. A fresh replay of copied archives completed 37 decoding checks on copied artifacts, including independent reconstruction of the full audio ellipse archive. A companion executable audit recomputes arithmetic and accounting identities.

Contribution and prior work

The contribution is the experimentally bounded comparison of these predictors under complete-byte accounting, including neutral and negative results. It is not the invention of predictive lossless coding, autoregressive prediction, residual coding, or description-length model selection. Rissanen’s minimum description length formulation evaluates a model together with the description it enables [1]. FLAC already combines prediction and lossless residual coding in a mature standardized format [2]. Better numerical prediction alone does not establish a smaller complete archive.

Measurement contract

The principal outcome is complete stored archive length in bytes, with headers, per-file descriptors, learned parameters, residuals, exceptions, and source wrappers included where applicable. Savings are (baseline bytes minus candidate bytes) divided by baseline bytes. The I16/I17 comparisons explicitly exclude decoder executable size for every method. “Independent verification” below denotes saved checks by separately implemented decoding or extraction logic, not independent third-party replication.

Phase pair coding and the neutral transfer control

Frozen phase pair mode

I12 augments the incumbent C8 CSV codec with one paying phase-pair mode and a C8 fallback. A previously decoded numeric column in [−1, 1] supplies x; a candidate target y is screened over the first min(4096, row count) rows for $|x^2 + y^2 - 1| \le 0.00001$ using decimal arithmetic. Names and hardcoded column identities do not select the relation. The source is rounded to scale 10^9 with ties to even. An integer floor square root of $10^{18} - X^2$ predicts target magnitude; the descriptor fixes target precision rounding. Sign, precision, exact integer residuals, and literal exceptions are stored. Brotli q11/w24 compresses the descriptor stream. The screen is a bounded eligibility test, not a universal relationship detector.

I12 complete archive	Bytes
C8 incumbent	10,223,513
C9 phase pair	9,437,726
Historical per-file Brotli control	12,401,316

The 42,215,215-byte development CSV has 262,144 rows and 21 columns. C9 saves 785,787 bytes (7.6861%) on already exposed data. Complete C9 accounting is 9,437,151 payload + 495 inner metadata + 80 outer header = 9,437,726 bytes. The phase target stream falls from 940,598 to 154,811 bytes. The Brotli control was retained from an earlier experiment rather than rerun in I12.

Saved fresh-process checks decoded both full archives without source access. A separate C# extractor and Python integer-square-root decoder reconstructed all 262,144 phase targets, including 139,799 nonzero residuals and 162 literal escapes. This independent check covers the new phase stream and accounting, not every inherited C8 mode. The exact source spellings are part of the losslessness contract. An older 178,056-byte estimate lacks a recovered executable and is not reproduced by I12.

I13 neutral bank

Separate CSV bank	Files	C8 bytes	C9 bytes
Voxel snapshots	45	3,673,781	3,673,781
Analysis tables	15	1,950,668	1,950,668
Total	60	5,624,449	5,624,449

The unchanged I12 implementation admitted no phase pair: 0 wins, 60 ties, and 0 losses. Source size totals 21,995,312 bytes; Brotli totals 6,233,936 bytes. Saved exact fresh decodes passed for all 180 archives. These files share project ancestry; 14 are header-only, and one 134-column file exceeds the frozen 64-column search limit. Equality therefore demonstrates fallback behavior on this bank, not general absence of phase structure. A verifier reflection-binding defect was corrected separately without changing measured archives or the frozen codec.

Transfer to an independent pendulum dataset

Dataset and exact object being compressed

I14 uses HumanCompatibleAI/ppo-Pendulum-v1 at revision b9bfc53d02f7616b2fd8c44d51ba72233d0672b9 [3]. The recorded observations are cosine of angle, sine of angle, and angular velocity, consistent with the Gymnasium observation definition [4]. The retained Parquet object contains 200 trajectories and 40,200 observations including endpoints. Each original float32 observation is exported with shortest round-trip spelling into episode, step, cos_theta, sin_theta, and angular_velocity columns. No normalization, recomputed trigonometry, row removal, or reordering is reported.

The exact target is the exported observation CSV, with a separate bitwise audit back to the original float32 values. It is not the original Parquet container and does not include reconstruction of actions, rewards, terminal flags, or information fields. The two precommitted episode ranges are halves of one external simulated dataset, not two independently sourced domains or a physical sensor validation.

Episode range	CSV bytes	C8 bytes	C9 bytes
0–99	825,235	147,359	124,153
100–199	846,406	150,415	125,758
Total	1,671,641	297,774	249,911

The unchanged I12 codec saves 23,206 and 24,657 bytes respectively: 47,863 complete bytes in total, or 16.0736% of C8. Brotli totals 382,002 bytes. In both halves, cosine predicts the magnitude of sine. The corresponding target streams change from 41,695 to 18,489 bytes and from 43,270 to 18,613 bytes.

Accounting and saved verification

The first C9 archive comprises 123,872 payload + 201 metadata + 80 outer bytes; the second comprises 125,477 + 201 + 80. Saved audits report 120,600 bit-exact float32 checks, independent reconstruction of all 40,200 phase targets, 34,539 nonzero residuals, 58 literal escapes, and six fresh source-free archive decodes.

This is the clearest positive transfer result for the unchanged phase-pair implementation: data published independently for reinforcement learning, selected because its observations expose the required relation. It remains a targeted transfer test on one dataset rather than a random sample of CSV workloads.

Availability boundary

A public dataset page is not itself a redistribution license. The retained dataset card contains no licensing declaration. This manuscript and companion packet include provenance, hashes, and reported measurements, but no redistributed dataset, private simulation export, or recording. A runnable archive replication release requires source artifacts and confirmed rights.

Ellipse prediction on Remembrance response tables

Input information and evaluation split

I16 constructs response tables from the first eight frames of five E42 Remembrance transport captures. Each row stores phase, velocity, coefficient vectors C and S, and response R alongside identifiers and flags. The ellipse predictor is Rhat = C cos(phi) + S sin(phi), with exact stored residuals preserving deviations and representation. Incoming velocity v is separately stored and is not part of this response target. Coefficients use recorded input-derived directional gain, geometry, and neighbor phase; targets independently sum recorded push tokens. The saved mutation control leaves coefficients unchanged when target pushes change.

J01 is pilot/training data. J02–J05 form the four-table evaluation total. I17 adds linear and quadratic comparisons retrospectively on these already known tables: J01 alone trains the models, and J02–J05 evaluate without subsequent tuning. This is not a new blind test of the full predictor family.

Table	C9	Ellipse	Linear	Quadratic
J01 training	140,118	134,231	147,968	148,791
J02	140,757	134,814	148,637	149,517
J03	140,903	135,027	148,818	149,769
J04	140,327	134,514	148,186	149,180
J05	138,872	132,840	146,529	147,471
Evaluation	560,859	537,195	592,170	595,937

Ellipse saves 23,664 bytes (4.2192%) against C9 across 6,741 evaluation rows. Phase permutation totals 589,200 bytes and zero prediction 588,589 bytes. The deterministic permutation rotates phase halfway within each frame while keeping geometry and targets fixed. It isolates phase placement within that construction, not a physical intervention or a statistical permutation test.

Model overhead and verification scope

Linear prediction uses an intercept and ten standardized inputs; quadratic prediction adds 55 products. Ridge strength is 10^{-6} with an unpenalized intercept. Per-archive binary64 models occupy 424 and 1,744 bytes. Both methods share base CSV, exact XOR residuals, byte-plane transpose, and Brotli q11/w24. For J02, quadratic residuals save 441 bytes versus linear but add 1,320 model bytes and one framing byte, leaving an 880-byte larger archive. Lower RMSE therefore fails to repay model overhead.

Saved I16 checks cover 15 ellipse/permutation/zero archives, 25,269 rational target components, and 85 recalculated coefficient groups; I17 adds ten exact linear/quadratic reconstructions. A SafeBridge process-boundary repair followed a damaged-payload hang without changing codec arithmetic or measured archives. No byte-identical historical source checkout was recovered. Recorded schema semantics and collection seals support provenance, not a replay of the original producer. I15’s earlier provenance obstruction remains unresolved.

A distinct causal audio adaptation

Predictor and controls

I18 adapts the geometric idea to a known mono 24-bit PCM recording at 96 kHz; it does not transfer the Remembrance codec unchanged. Audio lacks the recorded operator and phase supplied by those response tables. The ellipse predictor is $x[t] \approx a\,x[t-1] - x[t-2]$, constrained to $-2 \le a \le 2$. The quadratic quantity $x[t]^2 + x[t-1]^2 - a\,x[t]x[t-1]$ is invariant for the exact constant-coefficient recurrence and positive definite only for $|a| < 2$; endpoint cases are degenerate. Fitted, block-varying predictions with residuals do not establish a global physical ellipse.

Blocks contain 16,384 samples. Coefficients are fit using only the previous 2,048 decoded samples, with Q20 integer coefficients, exact integer normal equations, and ties away from zero. Controls are unrestricted causal two-lag prediction and fixed second differences (2, −1). All three share zigzag/Rice residual coding, exhaustive k = 0…27 selection, and raw24 fallback. Coefficients are regenerated from decoded history; the complete WAV wrapper is preserved.

Method	Complete bytes	Ellipse minus method
WavPack 5.9.0	16,052,448	708,899
FLAC 1.5.0	16,112,783	648,564
General two lag	16,753,016	8,331
Ellipse	16,761,347	0
Fixed second difference	16,842,241	−80,894

The source is 30,640,198 bytes, with 10,213,154 samples and a 736-byte wrapper. Ellipse saves 80,894 bytes (0.4803%) against fixed prediction, but loses against all three stronger controls. The registered requirement to beat both production codecs fails. This is a bounded negative result on one already exposed recording, not evidence that every geometric audio predictor must fail.

Complete accounting and checks

Ellipse accounting is 65 global header + 736 wrapper + 6,240 block headers + 16,754,306 padded payload = 16,761,347 bytes. Saved checks report exact full-WAV reconstruction for every method, independent integer/Rice reconstruction of all real samples and 81,920 synthetic samples, and matching logical bit counts and physical lengths. Production profiles were WavPack −hh −x6 with threads disabled and FLAC −8 −e −p with verification and foreign metadata retained. Single-run times are not a speed benchmark.

Later I19 integration into the known E07 bank saves only two bytes, 51,688,271 against 51,688,273, in one selected block. It does not rescue I18’s failed gate or establish practical benefit. I20 cross-recording work is a design only: not executed or implementation-frozen.

Interpretation and reproducibility

What the evidence supports

Complete-byte improvements survive in two bounded settings: a phase-pair mode on the development CSV and one external pendulum observation dataset, and operator-informed ellipse prediction on four simulation response tables. The neutral 60-file bank shows no gain when the mode admits no relation. The audio adaptation improves a weak fixed predictor but does not beat unrestricted two-lag prediction or production codecs. These are distinct algorithms and information contracts, not one frozen codec succeeding across all domains.

A small residual is not sufficient for a small archive. A geometric relation may leave sign, precision, spelling, or residual information undetermined. Exact preservation requires storing it. Likewise, quadratic prediction can lower error while increasing total bytes. These observations align with description-length accounting, but do not establish novelty of that principle, universal compression superiority, or any conclusion about prime-number algorithms.

Evidence status and next decisive test

This is a retrospective synthesis of recovered reports and receipts. The companion audit independently recalculates table totals, percentages, model-overhead differences, and complete-byte identities from transcribed values. It does not decode the original archives, verify their hashes against physical files, or certify the original encoders. Saved historical independent checks are described with their original scope. A fresh replay of copied archives completed 37 checks: I12 (2), I14 (6), I16 (15), I17 (10), and I18 (3 custom archives plus 1 independent ellipse decode). These are 37 checks, not 37 unique archives. Decodes used source-free interfaces, but filesystem access was neither restricted nor traced. Copied artifacts matched their originals and checked seals; originals were unchanged. This replay did not rerun encoders, I13, production audio decoders, the I14 export audit, I16 coefficient audit, or I17 independent decoder and RMSE audit.

The deliverable does not redistribute measured archives or frozen complete implementations, so recipients cannot reproduce corpus decoding from this packet alone. The next step is a rights-cleared artifact release with exact versions and independent checks for the remaining audit scopes. Missing artifacts remain inconclusive; a reconstruction mismatch fails the affected archive. Any new transfer claim requires a new precommitted dataset and frozen algorithm, not retuning on these results.

References

[1] J. Rissanen. Modeling by shortest data description. Automatica 14(5), 465–471, 1978. https://doi.org/10.1016/0005-1098(78)90005-5

[2] M. Q. C. van Beurden and A. Weaver. Free Lossless Audio Codec (FLAC). RFC 9639, December 2024. https://www.rfc-editor.org/rfc/rfc9639.html

[3] HumanCompatibleAI. ppo-Pendulum-v1. Pinned dataset revision b9bfc53d02f7616b2fd8c44d51ba72233d0672b9. https://huggingface.co/datasets/HumanCompatibleAI/ppo-Pendulum-v1

[4] Farama Foundation. Gymnasium Pendulum documentation. https://gymnasium.farama.org/environments/classic_control/pendulum/

Experiment record identifiers: I12 RESULTS; I13 RESULTS and AUDIT; I14 RESULTS and AUDIT; I16 RESULTS, PROTOCOL, and INDEPENDENT_VERIFY; I17 REPORT and VERIFICATION; I18 REPORT and INDEPENDENT_VERIFY; I19 REPORT; I20 DESIGN. These retained records are not publicly deposited. Their numerical transcriptions and missing-artifact requirements are supplied in the companion packet.

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